JOSH ONG

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SUMMARY

Third-year Interactive Media and Game Development student at SIT and DigiPen Singapore. Builds and debugs playable gameplay and AI features in C++ and Unreal Engine 5. Available for an internship from May 2027 to April 2028.

PROJECTS

Warbound — Gameplay Programmer, team of 9 | C++, OpenGL, ECS **2025-2026**

- Shipped playable builds across six graded milestones on a custom C++ engine
- Debugged and fixed gameplay issues across the 9 subsystems I owned (~5,700 lines): event bus, tile grid, turn-phase state machine, projectiles, 2D math
- Built a procedural dungeon generator with 4 level algorithms on 30x20 tile maps, seeded so any layout can be reproduced for testing
- Wrote a constrained spawn solver with flood-fill validation, so no generated level can strand the player
- Added JSON serialization so designers could save and reload generated layouts

HTN Planning in Unreal Engine 5 — AI Programmer, team of 3 **2026**

- Ported a hierarchical task network planner from a Godot 4 prototype to Unreal Engine 5 — 27 C++ source files, 9 primitive tasks
- Found and fixed a plan-lifecycle bug inherited from the original: a task that changed world state mid-execution caused the next plan's first step to be skipped; added plan versioning so a stale completion cannot mutate the live plan

Thanos World — Gameplay Programmer, team of 4 | C++, Alpha Engine **2025**

- Wrote the audio system: separate music and SFX channels under a master volume, a mute toggle that restores the current track, and no restart when two screens share the same music
- Built 80% of the power-up system: randomly spawned shield and infinite-energy pickups kept in a linked list, with pickup collision and memory cleanup

A* Pathfinding — Solo | C++ **2026**

- Implemented A* with 5 heuristics, rubberbanding and Catmull-Rom path smoothing
- Replaced the open list's linear scan with a binary heap, for O(log n) node selection

Terrain Analysis — Solo | C++ **2026**

- Built ten tactical analysis functions for AI agents, including influence maps with occupancy propagation and enemy field of view over a grid

Ruined City — Level Designer | Unreal Engine 5 **2026**

- Built a walking-simulator level with no combat, so light and framing do the guiding: inside buildings, one shaft of light through a broken roof marks the way on

EXPERIENCE / LEADERSHIP

Part-time Crew — Habitat Coffee / Hellu Coffee **Feb 2019 - Present**

- Served customers and handled the morning and afternoon rushes

Group Leader — Singapore Police Force **Jul 2022 - May 2024**

- Led a callsign of 3-4 officers responding to live incidents; briefed investigation officers on case handover

Business Analyst (Internship) — Accenture **Sep 2021 - Feb 2022**

- Queried and transformed large client datasets in SQL; scoped requirements with clients

Captain, Track and Field — Ngee Ann Polytechnic **Apr 2021 - Apr 2022**

- Led the team through the drills and workouts the coach set on training days, and worked with the EXCO to plan meetings and outings

Patrol Leader & Quartermaster — Scouts, Zhenghua Secondary **Jun 2017 - Feb 2018**

- Led a patrol of 7+ through each meeting's tasks, and taught knots and marching drills
- As quartermaster, ran the club and storage rooms and built an inventory list to track every item

EDUCATION

BSc Interactive Media and Game Development — DigiPen Singapore / SIT **Expected 2028**
Overseas Immersion Programme, Seattle: May - Jul 2026, classes overseas

Diploma, Electronic & Computer Engineering — Ngee Ann Polytechnic **2022**

SKILLS

Languages: C++, C, SQL · **Engines & tools:** Unreal Engine 5, Git, Figma · **Techniques:** ECS architecture, HTN planning, behavior trees, A* pathfinding, influence maps, procedural generation